

BOTA■ KOMPRESÖR & ÖLÇÜM ■STASYONU BORU ALIM I - ITP

Pipe Description: 24" (610 mm) x 12.70 mm, Grade X60 PSL 2, SAWH/LSAW (F=0.50 ■stasyon Borusu) | **Governing Standard:** BOTA■ 4-NGTL-0-GN-P-002-5120 Rev. 7 (82.5 Bar ■stasyon)

No	Inspection / Test Activity	Frequency & Extent	Sampling Location & Specimen	Test Method / Standard	Acceptance Criteria & Specified Limits	Governing Clause	Hold / Witness
1	Ladle Heat Analysis	Once per heat of steel	Ladle test sample	ASTM A751	C <= 0.16%, P <= 0.025%, S <= 0.010%, CE_IIW <= 0.40	BOTA■ 3.2.1	H/W
2	Product Chemical Analysis	Two analyses per test unit	Pipe body sample	ISO 14284	C <= 0.16%, P <= 0.025%, S <= 0.010%, CE_Pcm <= 0.22	BOTA■ 3.2.2.4	W
3	Pipe Body Tensile Test	2 sets per test unit (1 for IDARE 5 yrs)	Transverse body specimen	ISO 6892-1	Rt0.5 >= 415 MPa, Rm: 520-760 MPa, Af >= 21%, Y/T <= 0.90	BOTA■ 3.3.1.4	H/W
4	Weld Seam Tensile Test	2 sets per test unit	Transverse weld specimen	ASTM A370	Rm >= 520 MPa across weld, Elongation >= 10%	BOTA■ 3.3.2.1	W
5	CVN Impact Test (Body) at -20 °C	1 set (3 specimens) per lot at -20 °C	Transverse body Charpy V	ISO 148-1	At -20 °C: Min Avg 60 Joules, Min Individual 45 Joules	BOTA■ 3.3.5	H/W
6	CVN Impact Test (Weld & HAZ) at -20 °C	1 set weld + 1 set HAZ at -20 °C	Weld centerline and HAZ	ISO 148-1	At -20 °C: Min Avg 45 Joules, Min Individual 34 Joules	BOTA■ 3.3.5	H/W
7	Drop Weight Tear Test (DWTT)	Once per heat at 0 °C	Transverse DWTT specimen	API RP 5L3	At 0 °C: Average shear area >= 85%, No single specimen < 60%	BOTA■ 3.3.6.4	H/W
8	Residual Stress Ring Test	Every heat (per heat mandatory)	150 mm ring opposite weld	BOTA■ 3.3.9	S <= 0.10 x SMYS (Max 41.5 MPa)	BOTA■ 3.3.9	H/W
9	Hydrostatic Pressure Test	Each pipe (100% all pipes)	Full pipe length	BOTA■ 8.4	Pressure: 100% SMYS (170 bar), Duration: MINIMUM 20 SECONDS	BOTA■ 8.4.1	H/W
10	Weld Seam 100% NDT (UT + RT)	100% full length of weld	Online UT + Offline RT	ISO 10893-11/6	100% Online UT inspection plus RT verification of ends	BOTA■ 8.8.4	H/W
11	Pipe Ends UT Laminar Testing	100% of all pipe ends	50 mm band on pipe ends	ISO 10893-8	100% pipe ends scanned (min 50 mm band, no defect)	BOTA■ 8.8.4.4.2	W
12	Dimensional & Wall Thickness Verification	100% of all pipes (D, t, L)	Pipe body and pipe ends	BOTA■ 5.1/5.2	End ovality <= 50% API 5L Table 10; Straightness <= 0.10% L; t=12.70 mm (-8/+15%)	BOTA■ 5.1 & 5.5	W
13	Visual Surface Inspection	100% internal and external	Full body visual	BOTA■ 8.1.2	100% visual inspection, no crack, no sliver, no lamination	BOTA■ 8.1.2	W